

Project Milestone 4: Final Results and Project Report

Sales and Profit Analysis for Retail Decision Support Using the Global Superstore Dataset

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1. Introduction

Retail organizations operate in highly competitive environments where pricing and promotional strategies play a crucial role in performance. While discounting is commonly used to stimulate demand and increase sales volume, its impact on profitability is more complex. Decisions that prioritize revenue growth without considering margin effects can lead to inconsistent financial outcomes and reduced overall performance.

This project focuses on analyzing transaction-level retail data to better understand the factors that influence profitability. Using the Global Superstore dataset, the study explores how variables such as sales, discount, quantity, product category, and region interact to affect profit outcomes. The analysis moves beyond descriptive insights by applying predictive modeling techniques to evaluate whether profit can be estimated reliably and whether transactions can be classified as profitable or loss-making in advance.

By combining exploratory data analysis with regression and classification models, this project aims to provide a structured, data-driven approach to support pricing and discount decisions. The findings are intended to contribute to more informed decision-making by highlighting key risk factors and identifying patterns that may not be immediately visible through traditional analysis.

2. Research Questions

RQ-1 Which product categories, sub-categories, discounts, and quantities are most strongly associated with higher profit?

RQ-2 How accurately can future profit be predicted from historical sales data such as discount, quantity, category, and region?

RQ-3 Can sales transactions be classified as profitable or non-profitable in advance to support discount and pricing decisions?

Contribution:

This study contributes to existing research by:

- Shifting focus from revenue to profitability
- Combining regression and classification approaches
- Incorporating baseline comparison and validation
- Providing a decision-support perspective for retail pricing

3. Literature Review and Integration

Predictive analytics has been widely used in retail for demand forecasting and revenue optimization. However, most studies focus on sales rather than profit.

Research suggests:

- Discounts increase demand but reduce margins
- Revenue growth does not guarantee profitability
- Predictive models should support decision-making, not just accuracy

This study builds on existing literature by focusing on profit risk prediction at the transaction level. The findings from the analysis align with prior research, particularly in highlighting the negative impact of discounting on profitability.

4. Dataset Description

This project uses the Global Superstore dataset, which contains transaction-level retail data from a global company. Each row represents a single product sale and includes details such as sales amount, profit, discount, quantity, and product category. The dataset consists of 51,290 transactions and 27 variables, including both numerical and categorical features. For this analysis, the most important variables are:

- **Sales** - revenue generated from each transaction
- **Profit** - profit or loss from the transaction
- **Discount** - discount applied
- **Quantity** - number of units sold
- **Category and Sub-Category** - product classification

These variables help in understanding what drives profitability in retail transactions.

Descriptive Statistics

Initial exploration shows that:

- Sales values vary widely, indicating both small and large transactions.
- Profit includes both positive and negative values, meaning some sales result in losses.
- Discounts are not applied uniformly and vary across transactions.

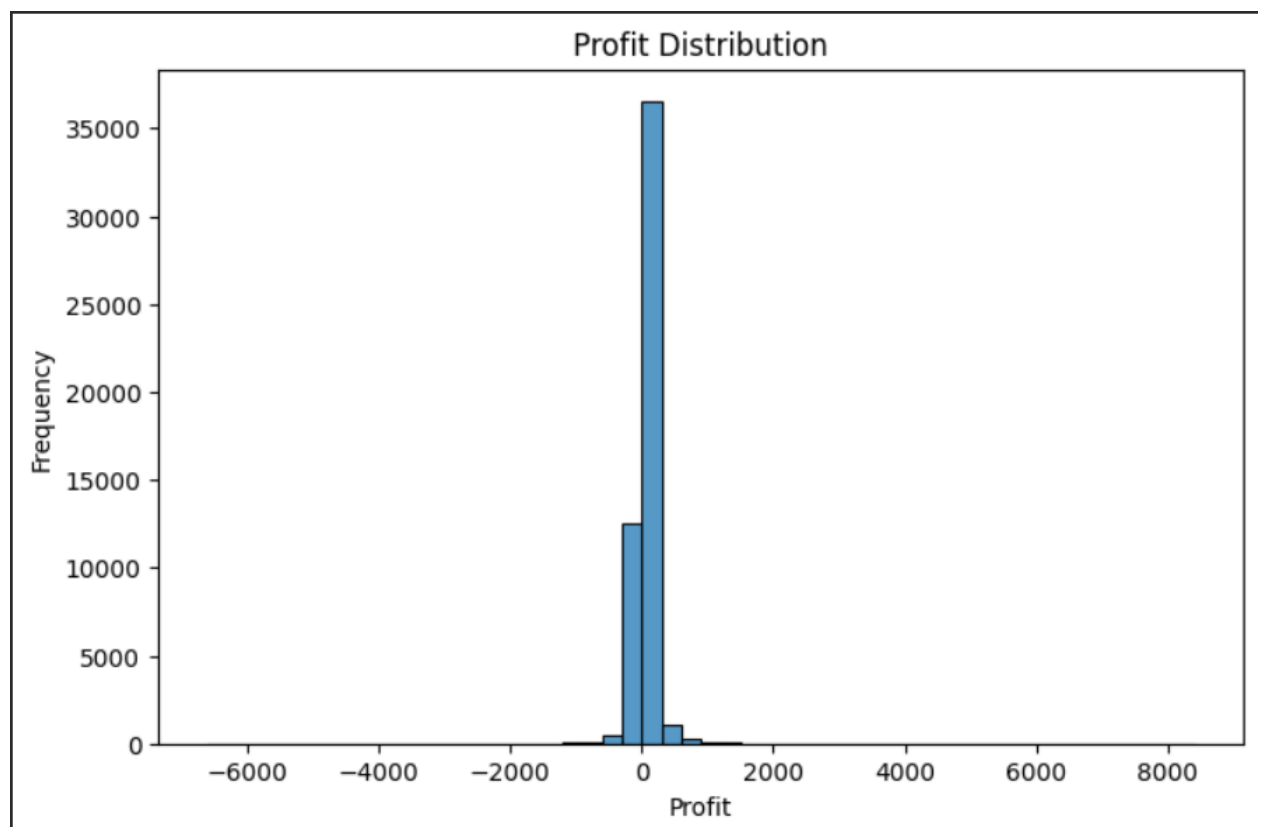
The dataset contains no missing values, which makes it clean and reliable for modeling without major preprocessing.

Data Visualization and Patterns

Several visualizations were developed to explore relationships within the dataset:

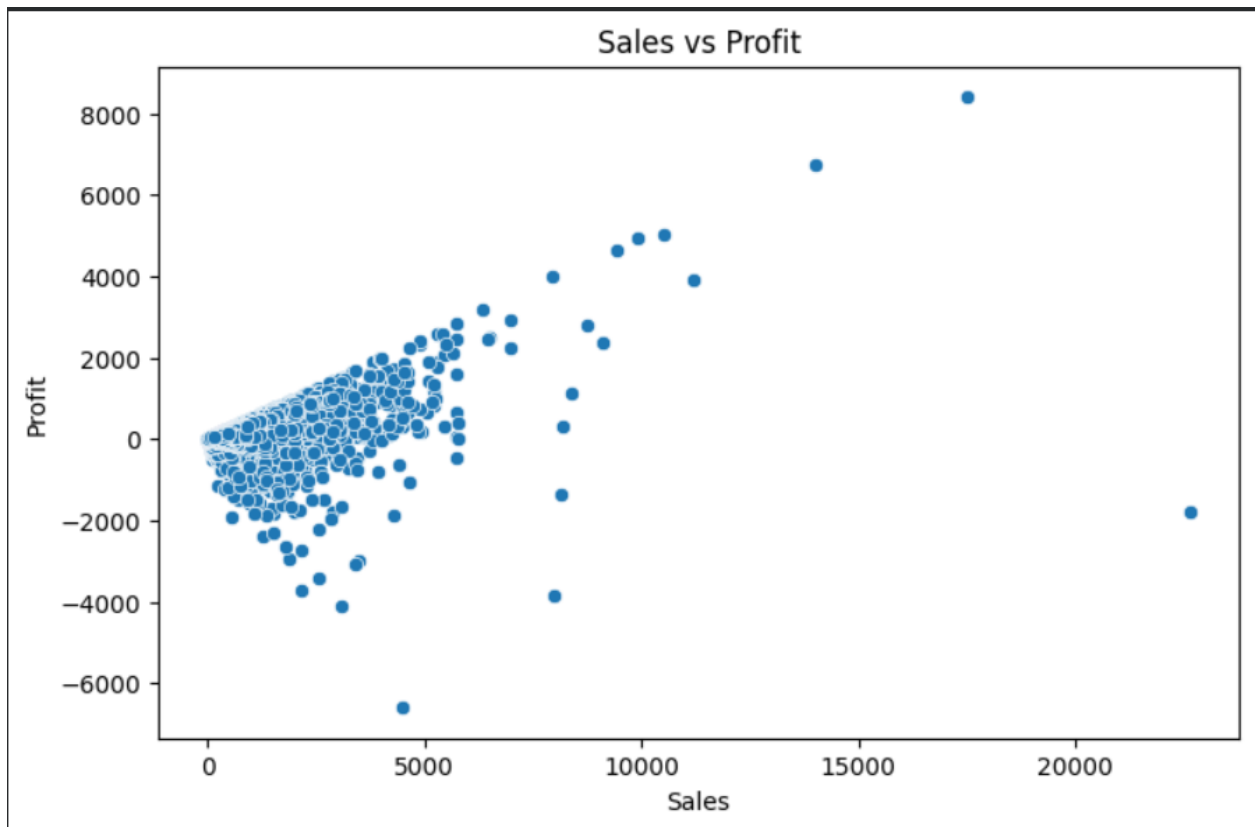
- **Profit Distribution:**

As shown in Figure 1, the distribution of profit is skewed, with most transactions generating small profits and some resulting in large losses or gains. This variability suggests that profit is influenced by multiple interacting factors.



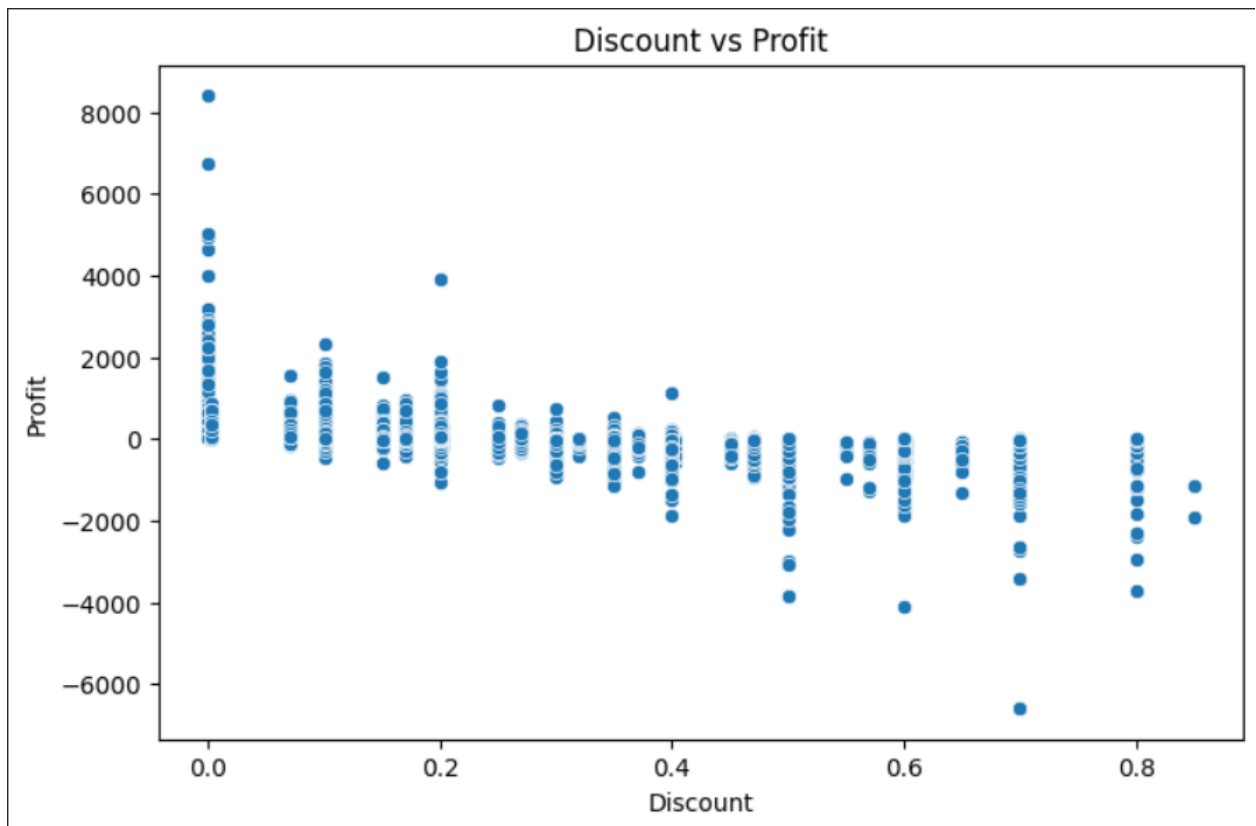
- **Sales vs Profit:**

The scatter plot shows that higher sales do not necessarily lead to higher profit. This indicates that revenue alone is not a reliable indicator of profitability.



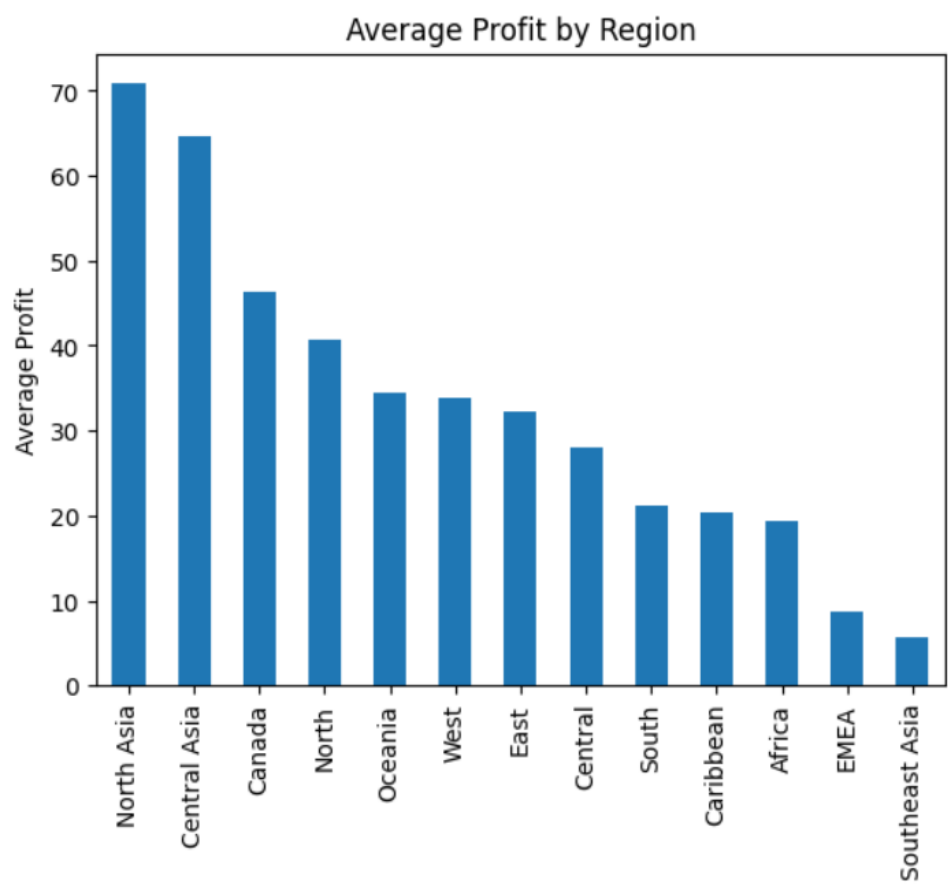
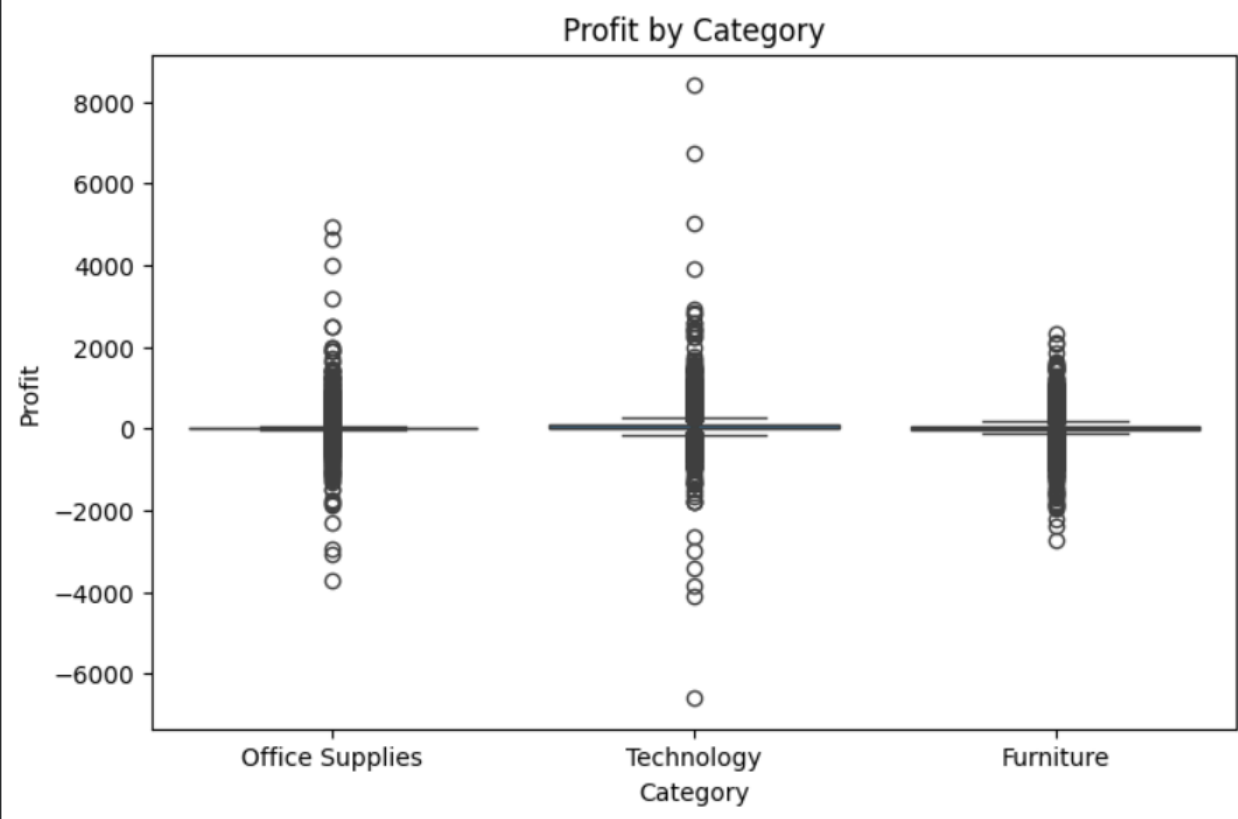
- **Discount vs Profit:**

A clear negative relationship is observed between discount and profit. As discounts increase, profit tends to decrease. This is one of the most important findings in the analysis.



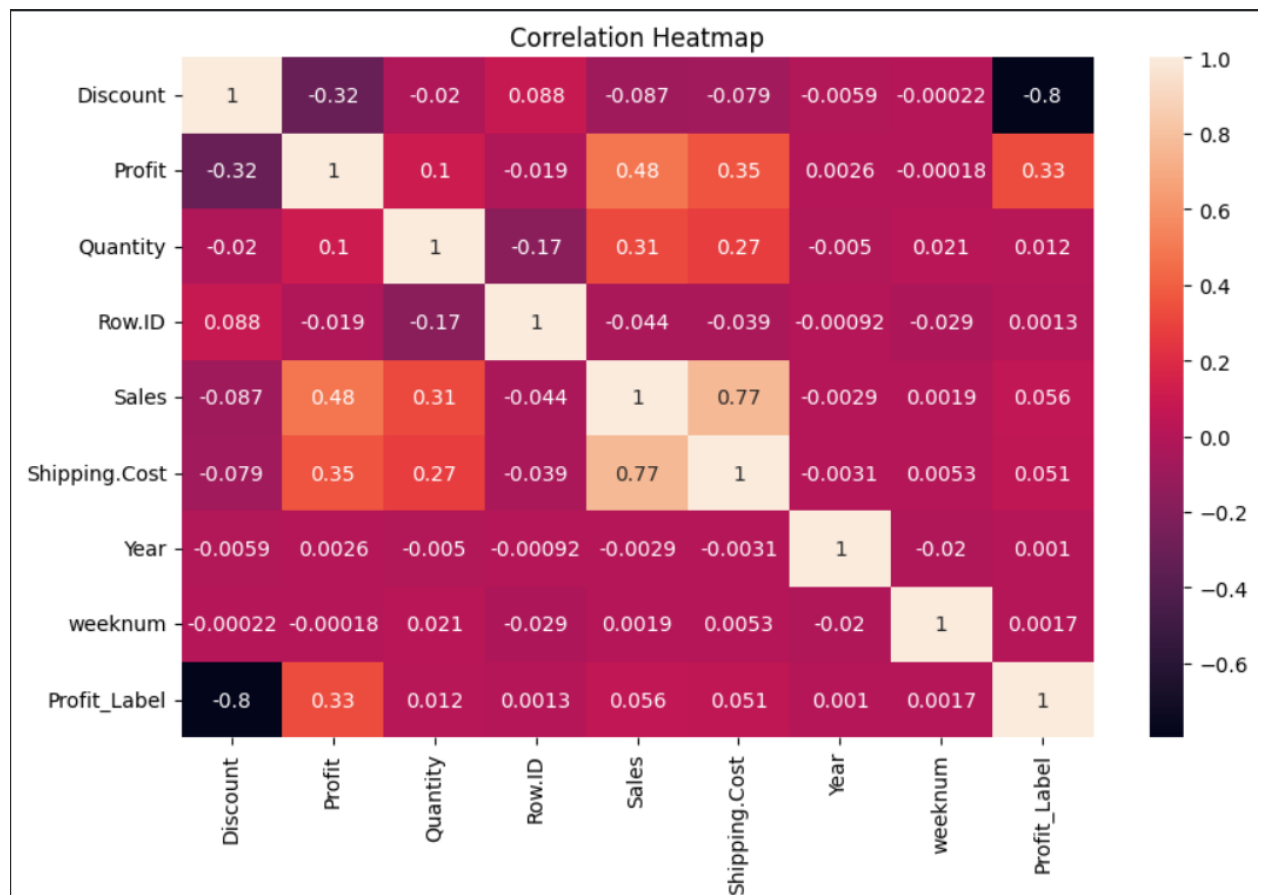
- **Profit by Category and Region:**

Profit varies across product categories and regions. Technology products generally generate higher profit compared to other categories. Regional variation suggests that location-specific factors may influence performance.



- **Correlation Heatmap:**

The heatmap shows that discount has a negative correlation with profit. However, these relationships represent correlation, not causation, and should be interpreted cautiously.



Key Insights

The exploratory analysis reveals the following:

1. Profitability varies significantly across transactions
2. Discount has a strong negative association with profit

3. Sales alone does not guarantee profitability
4. Nonlinear relationships are likely present in the data

These observations justify the use of predictive models to support pricing and discount decisions.

5. Data Preparation

Data Cleaning

The dataset was reviewed for missing values and duplicates. The dataset contains no significant missing values, making it suitable for modeling without major preprocessing. An unnecessary column (记录数) was removed as it did not contribute meaningful information

Feature Selection

For initial modeling, the following variables were selected:

- **Input features:** Sales, Discount, Quantity
- **Regression target:** Profit
- **Classification target:** Profit_Label

These features were chosen because they directly influence revenue and cost. Categorical variables such as Category and Region were excluded in the initial model to avoid complexity from encoding and to maintain interpretability. These variables were selected because they directly influence transaction-level revenue and profitability and are available without requiring complex preprocessing.

Data Transformation

A binary classification variable, Profit_Label, was created to support classification modeling:

- Profit > 0 = Profitable (1)
- Profit ≤ 0 = Not Profitable (0)

This transformation allows profitability to be analyzed as a classification problem.

Train-Test Split

An 80/20 split was used to balance training data availability and evaluation reliability. The training data was used to build the models, while the testing data was used to evaluate performance on unseen data.

6. Model Evaluation

6.1 Regression Models

Baseline Regression:

- **RMSE:**171.43
- **R²** :0.00

This indicates that simply predicting the mean provides a weak benchmark.

Linear Regression:

Linear Regression was implemented as a baseline model to predict profit. Model performance was evaluated using:

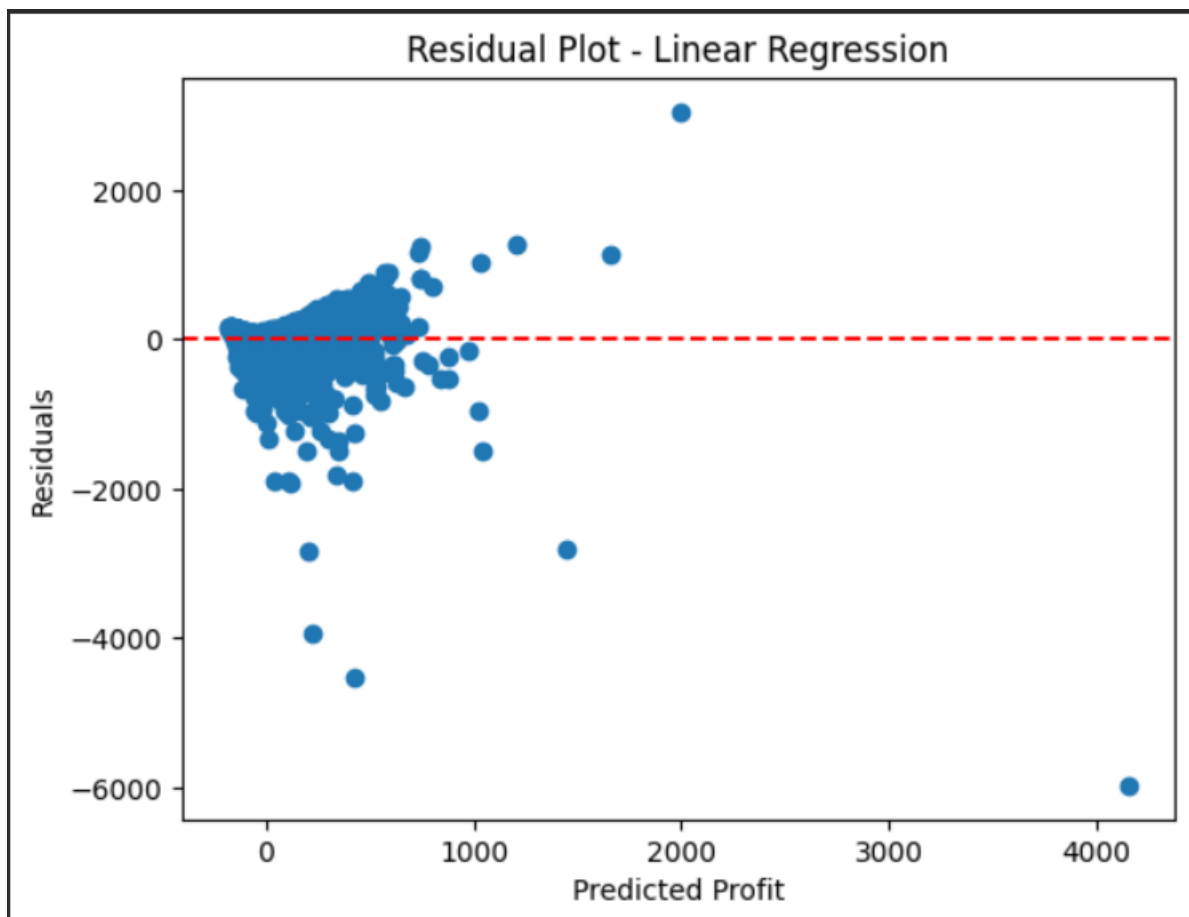
- Root Mean Squared Error (RMSE): Measures prediction error
- R^2 Score: Indicates explained variance

The model achieved:

- **RMSE:** 156.32
- **R^2 :** 0.17

The low R^2 indicates that the model explains only 17% of the variance in profit. This suggests that linear relationships alone are insufficient to capture the complexity of the data.

Residual patterns also indicate potential violations of assumptions such as constant variance.



Decision Tree Regression:

A Decision Tree model was used to capture nonlinear relationships between features and profit.

The model achieved:

- **RMSE:** 120.26
- **R²:** 0.51

The improved performance suggests that profit is influenced by nonlinear interactions between variables.

To reduce overfitting, tree depth was controlled.

This highlights the importance of selecting appropriate modeling techniques when dealing with complex, real-world business data.

6.2 Classification Model

Baseline Classification

- Accuracy: 0.74

Logistic Regression:

A Logistic Regression model was developed to classify transactions as profitable or not. The model achieved good accuracy and was able to correctly identify a large proportion of profitable transactions.

- Accuracy: **0.9185**
- Precision:**0.9195**
- Recall:**0.9754**
- F1-score:**0.9466**

6.3 Confusion Matrix Analysis

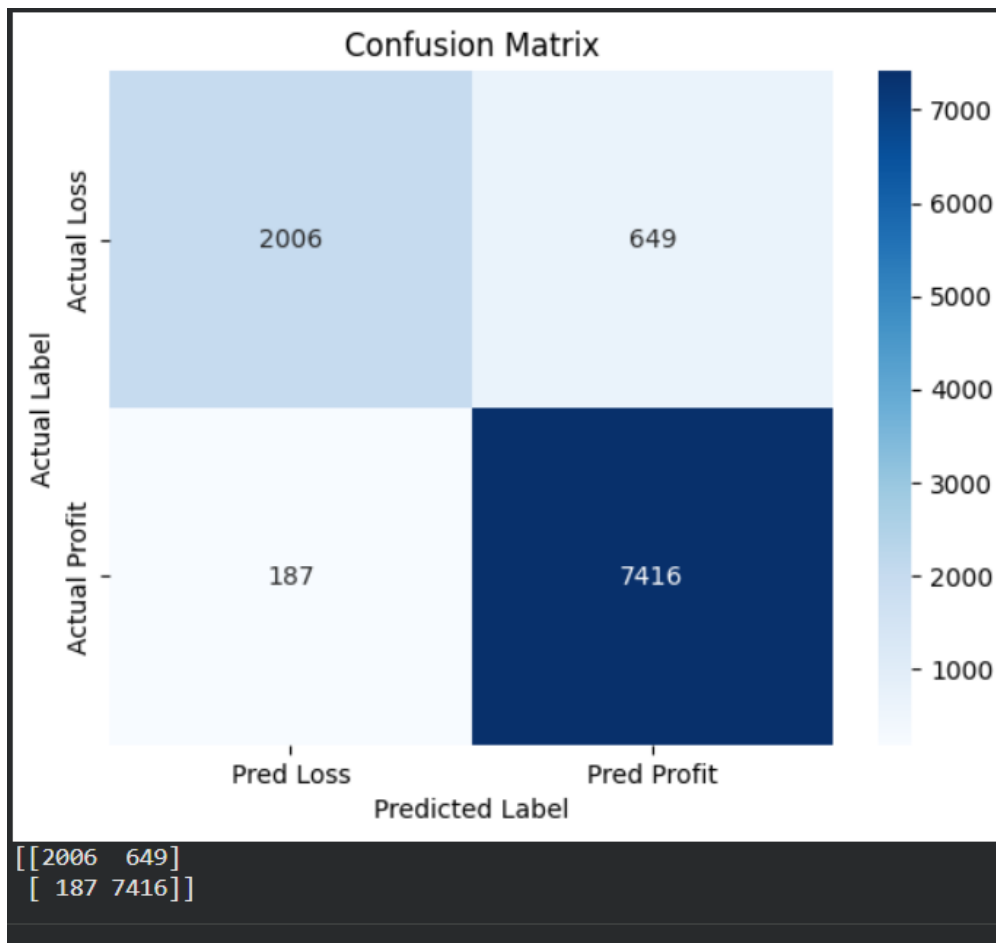
As shown in Figure 7, the confusion matrix provides a detailed breakdown of classification performance.

The model performs strongly, with high accuracy and recall. However, results may still be influenced by class distribution and should be interpreted carefully.

Confusion matrix results:

- False Positives: predicted profit but actual loss
- False Negatives: missed profitable transactions

False positives are more critical because they may lead to poor pricing or discount decisions.



6.4 Error Analysis

The regression scatter plot shows difficulty in predicting extreme values, indicating limitations in handling outliers.

7. Model Comparison

Model	Strength	Weakness
Linear Regression	Simple & interpretable	Low explanatory power
Decision Tree	Captures nonlinear patterns	Risk of overfitting
Logistic Regression	Strong classification	Limited feature set

8.Overall Findings

The results of the initial models indicate:

- The Decision Tree model outperforms Linear Regression, showing that profit relationships are nonlinear.
- Discount is a key factor influencing profitability.
- Regression models show moderate predictive power, while classification models perform strongly.
- Results should be used cautiously for decision-making
- Profit varies across categories and regions
- Classification helps identify risky transactions

Overall, these results demonstrate that predictive models can be useful tools for improving retail decision-making.

Profitability is influenced by multiple interacting factors and cannot be explained using a simple linear model. While regression models show moderate predictive ability, classification models perform strongly.

The predictive models provide useful insights but should be interpreted as decision-support tools rather than exact predictors.

9. Limitations and Ethical Considerations

Limitations

- Limited feature set
- No causal inference
- Model assumptions may not hold
- Risk of overfitting
- Missing cost and customer data

Ethical Considerations

- Risk of biased decision-making
- Potential misuse of predictions
- Need for transparency

10. Future Work and Recommendations

Future Work

- Include cost and customer-level data
- Use advanced models
- Improve feature engineering
- Apply time-based validation

Business Recommendations

- Avoid excessive discounting
- Use predictive models to support pricing decisions
- Monitor high-risk transactions

11. Conclusion

This project demonstrates that predictive analytics can support profitability-focused decision-making in retail. While regression models show moderate performance, classification models provide strong predictive capability. However, further improvements are required before real-world deployment. This highlights the importance of combining data-driven insights with managerial judgment in real-world decision-making.

12. Code Documentation

All analysis and modeling were implemented in Python using Jupyter Notebook.

The project is organized within a GitHub repository, which includes:

- Jupyter Notebook containing data analysis and models
- Dataset used for the project
- PDF version of the report
- README file describing the project structure and stages

GitHub Repository:

<https://github.com/sahanashaikh14/global-superstore-profit-analysis>

13. Video Presentation

A five-minute video titled “**Initial Results and Code**” was created using Zoom. The video has been uploaded and is accessible through the GitHub repository as part of the project submission.

<https://github.com/sahanashaikh14/global-superstore-profit-analysis/blob/98929208ab505085a9827a924e23af7453cf7756/Milestone%203%20Video%20Presentation.mp4>

The presentation includes:

- Overview of the dataset and objectives
- Key findings from exploratory data analysis
- Explanation of preprocessing steps
- Walkthrough of regression and classification models
- Summary of results and insights

14. References

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